Codominance & Sex-Linked Characteristics

Exam code: 0610 & 0970

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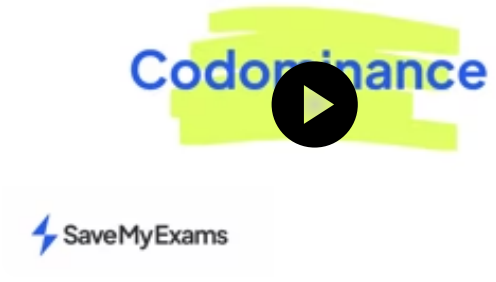

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Help me understand

Codominance: Extended

Extended Tier Only

Codominance

- Codominance occurs when **both alleles in heterozygous organisms contribute** to the phenotype
- Inheritance of blood group is an example of codominance

- There are three alleles of the gene governing this instead of the usual two
- Alleles I^A and I^B are codominant, but both are dominant to I^O
- I represents the gene and the superscript A, B and O represent the alleles
- I^A results in the production of antigen A in the blood
- I^B results in the production of antigen B in the blood
- I^O results in no antigens being produced in the blood
- These three possible alleles can give us the following genotypes and phenotypes:

Stuck on something?

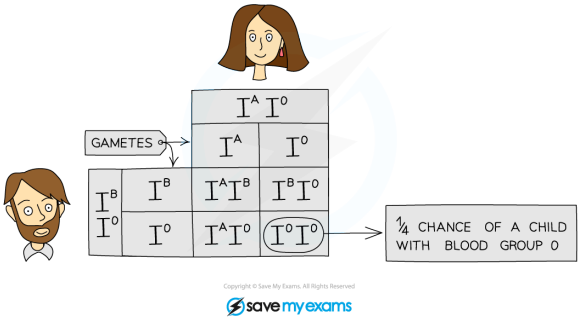
Ask questions and get help with exactly what you're revising.

Got it

$I^A I^A$ OR $I^A I^O$	A
$I^B I^B$ OR $I^B I^O$	B
$I^A I^B$	AB
$I^O I^O$	O

- We can use genetic diagrams to predict the outcome of crosses that involve codominant alleles:

'Show how a parent with blood group A and a parent with blood group B can produce offspring with blood group O'



Punnett square showing the inheritance of Blood Group

- The parent with blood group A has the genotype $I^A I^O$
- The parent with the blood group B has the genotype $I^B I^O$
- We know these are their genotypes (as opposed to both being homozygous) as they are able to produce a child with blood group O and so the child must have inherited an allele for group O from each parent
- Parents with these blood types have a 25% chance of producing a child with blood type O

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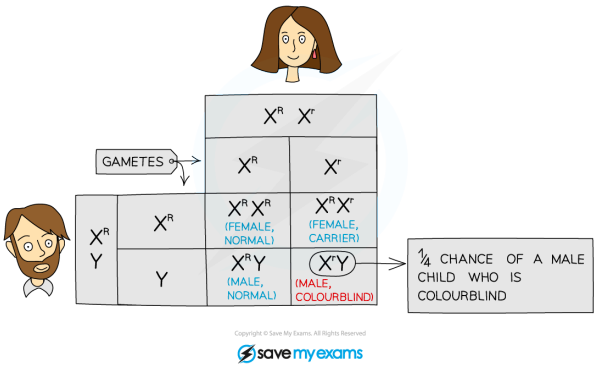
Characteristics: Extended

Extended Tier Only

- Alleles on the same chromosome are said to be **linked**
- When alleles that control a particular characteristic are found on the **sex chromosomes**, we describe the inheritance that results as **'sex linked'**
- In almost all cases, there are **only alleles on the X chromosome** as the **Y chromosome is much smaller**
- Because males **only have one X chromosome**, they are **much more likely to show sex-linked recessive conditions** (such as red-green colour blindness and haemophilia)
- Females, having two copies of the X chromosome, are likely to inherit one dominant allele that masks the effect of the recessive allele
- A female with one recessive allele masked in this way is known as a **carrier**; she doesn't have the disease, but she has a 50% chance of passing it on to her offspring
- If that offspring is a male, he will have the disease
- The results of a cross between a normal male and a female who is a carrier for colourblindness is as follows:



Punnett square showing the inheritance of colourblindness, an X-linked condition

- In the cross above, there is a 25% chance of producing a male who is colourblind, a 25% chance of producing a female carrier, a 25% chance of producing a normal female and a 25% chance of producing a normal male

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